

Dr. Doc:

 #2017 RESPIRATORY PHARMACOLOGY EXAM

 In comparison to inhaled β_2 -agonists, the inhaled anticholinergics:

- A. Are more effective in bronchial asthma
- B. Are better suited for control of an acute attack of asthma
- C. Produce slower response in bronchial asthma
- D. Produce rapid response in bronchial asthma

 Answer: C. Produce slower response in bronchial asthma

 Explanation:

Short-acting β_2 -agonists (SABAs), such as salbutamol, have a rapid onset and are important bronchodilators for acute symptom relief.

Anticholinergics such as ipratropium also produce bronchodilation, but their onset is generally slower than inhaled β_2 -agonists and they are particularly important in COPD.

 Remember:

SABA → rapid bronchodilation ⚡

Ipratropium → slower bronchodilation 🌱

 A 54-year-old man was diagnosed with status asthmaticus. Which of the following is the most preferred drug to treat this problem?

- A. IV theophylline
- B. IV methylprednisolone
- C. Nebulized ipratropium
- D. Nebulized salbutamol

✅ Answer: B. IV methylprednisolone

💡 Explanation:

🚨 Status asthmaticus

This refers to a severe, persistent asthma exacerbation that does not respond adequately to initial treatment and can become life-threatening.

The major underlying problem is severe:

🔥 Airway inflammation

🫁 Bronchospasm

💧 Airway edema

🌀 Mucus obstruction

Systemic corticosteroids are essential in severe exacerbations because they rapidly address the underlying inflammatory process.

💊 Methylprednisolone

IV methylprednisolone → powerful systemic anti-inflammatory effect

Other treatments, including nebulized salbutamol and ipratropium, may also be used during severe exacerbations for bronchodilation.

★ Important distinction:

Salbutamol → rapidly opens the airway ⚡

Systemic corticosteroid → suppresses severe inflammation 🔥

🧬 A patient with severe persistent asthma received combination therapy with high-dose ICS + LABA but failed to respond and has raised IgE. This case might be treated by:

- A. Omalizumab
- B. Systemic corticosteroid
- C. Mepolizumab
- D. Tiotropium

✅ Answer: A. Omalizumab

💡 Explanation:

This patient has:

🔥 Severe persistent asthma

🔴 Failed high-dose ICS + LABA

 Elevated IgE

 Likely allergic phenotype

This points toward:

★ Omalizumab

▼ Omalizumab is a monoclonal antibody directed against IgE.

⚙ Mechanism

Omalizumab binds free IgE



↓ IgE available to bind FcεRI receptors on mast cells/basophils



↓ allergic mediator release



 ↓ allergic airway inflammation

🧠 IMPORTANT DIFFERENTIATION

Omalizumab

 Anti-IgE

Mepolizumab

 Anti-IL-5

 One of the following drugs is effective in the treatment of aspirin-induced asthma:

- A. Zafirlukast
- B. Salbutamol
- C. Mepolizumab
- D. Pholcodine

 Answer: A. Zafirlukast

 Explanation:

Aspirin-exacerbated respiratory disease is associated with increased production of cysteinyl leukotrienes.

▼ Aspirin/NSAID exposure inhibits cyclooxygenase activity, causing arachidonic acid metabolism to shift toward the 5-lipoxygenase pathway.



↑ Leukotrienes



Bronchoconstriction + nasal symptoms

 Zafirlukast

Zafirlukast is a:

▼ CysLT₁ receptor antagonist

➡ It blocks the effects of cysteinyl leukotrienes.

Therefore it can be useful in aspirin-induced asthma.

👉 Dextromethorphan is preferred to codeine as an antitussive because:

- A. Less risk of dependence & respiratory depression
- B. Acts peripherally, not centrally
- C. It has potent analgesic effect while codeine is a selective antitussive
- D. Less constipating
- E. A and D

✅ Answer: E. A and D

💡 Explanation:

❌ Both drugs act centrally to suppress the cough reflex, so option B is incorrect.

● Dextromethorphan

 It is a nonopioid antitussive and has:

 Less risk of respiratory depression at therapeutic doses

 Much less constipating effect

 Less opioid-related dependence potential

 Codeine

 Codeine is an opioid and can cause:

 CNS depression

 Respiratory depression

 Constipation

 Dependence/misuse potential

 Exam pearl:

 Dextromethorphan = central antitussive + nonopioid + minimally constipation

🧠 Which of the following is NOT correctly matched with its use?

- A. Dextromethorphan → Antitussive
- B. Codeine → Antihistamine
- C. Phenylephrine → Nasal decongestant
- D. Guaifenesin → Expectorant

✅ Answer: B. Codeine → Antihistamine

💡 Explanation:

Let's match them correctly:

🧠 Dextromethorphan

→ Nonopioid antitussive

💊 Codeine

→ Opioid antitussive

👉 Phenylephrine

→ α -adrenergic agonist / nasal decongestant

💧 Guaifenesin

→ Expectorant

Therefore:

❌ Codeine is NOT an antihistamine.

★ Memory:

Codeine → cough suppressant 🗣️

Dextromethorphan → cough suppressant 🗣️

Guaifenesin → mucus/expectorant 💧

Phenylephrine → nasal decongestant 👉

🧬 Which medication inhibits 5-lipoxygenase and consequently inhibits the formation of leukotriene B₄ and cysteinyl leukotrienes?

- A. Cromolyn
- B. Zafirlukast
- C. Zileuton
- D. Albuterol

✅ Answer: C. Zileuton

💡 Explanation:

The arachidonic acid pathway can be summarized as:

Arachidonic acid



5-lipoxygenase (5-LOX)



Leukotrienes Including:

LTB₄, LTC₄, LTD₄, LTE₄

💊 Zileuton

Zileuton = 5-LOX inhibitor 🚫

Therefore:

Zileuton → ↓ 5-LOX → ↓ LTB₄ + ↓ cysteinyl leukotrienes

vs Zafirlukast

Zafirlukast does not inhibit 5-LOX.

🚫 It blocks the CysLT₁ receptor.

🧠 Memory:

Zileuton → blocks synthesis 🏭

Zafirlukast → blocks receptor 🚫

🎤 A drug used in asthma that induces hoarseness of voice and candidiasis is:

A. Ipratropium

B. Fluticasone

C. Salbutamol

D. Salmeterol

✅ Answer: B. Fluticasone

💡 Explanation:

Fluticasone is an inhaled glucocorticoid (ICS).

Important local adverse effects include:

🎤 Dysphonia → hoarseness of voice

🍄 Oropharyngeal candidiasis → oral thrush

These occur because corticosteroid deposition in the mouth and throat can locally suppress immune responses.

Prevention

Using a spacer when appropriate and rinsing the mouth after inhalation can reduce the risk of oral candidiasis.

Exam pearl:

▼ ICS → hoarseness + oral candidiasis

 All of the following statements regarding theophylline are TRUE EXCEPT:

- A. It inhibits phosphodiesterase enzyme & increases cyclic AMP in mast cells
- B. It works in part by blocking adenosine A₁ receptors
- C. If given with rifampicin, decreased levels of theophylline may occur
- D. If given with erythromycin, decreased levels of theophylline may occur

 Answer: D. If given with erythromycin, decreased levels of theophylline may occur

Explanation:

Theophylline has several actions:

Mechanisms

Inhibits phosphodiesterase (PDE) → ↑ cAMP

Blocks adenosine receptors, including A₁

Produces bronchial smooth-muscle relaxation

🎯 Theophylline is extensively metabolized in the liver, with CYP1A2 being an important enzyme.

● Rifampicin

Rifampicin is an enzyme inducer.

Rifampicin → ↑ metabolism → ↓ theophylline concentration

Therefore, C is true.

● Erythromycin

✓ Erythromycin can inhibit theophylline metabolism.

Erythromycin → ↓ metabolism → ↑ theophylline concentration → ↑ toxicity

Therefore, it does not decrease theophylline levels.

★ Memory:

Inducer → ↓ drug level

Inhibitor → ↑ drug level

All of the following are side effects of β_2 -agonists EXCEPT:

A. Tachycardia ❤️

B. Hypokalemia ↓K⁺

C. Hyperkalemia ↑K⁺

D. Hyperglycemia 🩸

✅ Answer: C. Hyperkalemia

💡 Explanation:

β_2 -agonists can cause:

❤️ Tachycardia

🩸 Hyperglycemia

↓K⁺ Hypokalemia

↓Mg²⁺ Hypomagnesemia

β_2 stimulation promotes movement of K^+ from the extracellular fluid into cells, causing:

β_2 stimulation $\rightarrow$ intracellular K^+ shift $\rightarrow$ $\downarrow$ serum K^+ $\rightarrow$ hypokalemia

Therefore, hyperkalemia is NOT a typical adverse effect.

★ Exam pearl:

✓ β_2 agonists $\rightarrow$ Hypo- K^+ , not Hyper- K^+ .